

HERMANN PAUWELLS RIVERA

Full-Stack Developer | Software Engineer

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PROFILE

Full-stack developer specialized in TypeScript, React/Next.js and Node.js, graduating in December 2026 with a Computer Science Engineering degree from Tecnológico de Monterrey. Experienced building software for real businesses and users, including e-commerce, real-time systems and APIs. Additional experience in C++/JUCE, DSP and machine learning.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL, Swift, C#

Frontend: React, Next.js, Vite, Tailwind CSS, Three.js

Backend & Data: Node.js, Express, Supabase, PostgreSQL, MySQL, FastAPI, REST, SSE

Tools: Git, GitHub, Docker, Vercel, GitHub Actions, CI, JUCE, TensorFlow, Unity

EXPERIENCE

Full-Stack Developer (Freelance) — Juguetería El Arbolito

2026–present

- Developing and implementing an e-commerce and admin platform for a single-branch retail business (catalog, cart, checkout, backend).
- Migrated a 2,395-product catalog from the legacy point-of-sale system (Eleventa) into PostgreSQL/Supabase.
- Implementing Mercado Pago payments and inventory synchronization between online sales and the in-store point of sale.

SELECTED PROJECTS

Lumina Reservations (WorkHub MTY)

Academic team project · Node/Express backend, real-time (SSE) and frontend

2026

- Implemented the reservation backend in Node/Express (services, repositories, controllers, migrations) and real-time occupancy updates over SSE.
- Built the floor map and the reservation flow in React/Vite (dashboard, filters, confirmation) with role controls.

RainLab Northsky

Interactive 3D web application · Three.js

2026

- Interactive viewer with animated camera transitions, component views and a Blender/GLB asset pipeline.
- Reduced rendering draw calls from 637 to 127 (80%) through mesh batching; rendering pauses when the viewer is off screen.

folk-park

Personal project

2026

- Wavetable VST3/Standalone synthesizer in C++20 and JUCE with deterministic MIDI generation.
- DSP architecture and automated tests; validated with pluginval strictness 5.

HeatShield

iOS app

2025

Winner, Sustainability category, Swift Challenge Fest 2025 (Tecnológico de Monterrey)

- iOS app with real-time heat alerts (OpenWeatherMap), heat index and shelter maps.

Other projects: CrediFiel Challenge (Datathon 2025) · Multi-agent warehouse simulation (RL) · MAWI · VisualControl (IoT) · Mildred Pierce · GateGenius (HackMTY 2025).

EDUCATION

Computer Science Engineering (ITC) — Tecnológico de Monterrey, Monterrey Campus — 2021–2026 ·

Graduation: December 2026 · Available full-time from January 2027

ACTIVITIES

- SwiftTec Coding Community (2025) — hackathons and innovation challenges.
- Music production and sound design (FL Studio).

Spanish — native · English — C1